

## 6.4.1 – 6.4.2 Interviews and Surveys

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے...

CS XI

Chapter 6: Digital Literacy

Lecture 04 of 14

★ Golden

### 6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

#### 6.4.1 Interviews

##### LEARN IT

An **interview** is a **planned conversation** used to gather information **directly from a person**.

- It may be **one-to-one** (you and one person) or in a **small group**.
- Questions are **planned in advance**, so you do not forget anything important.
- Answers are usually **qualitative** (opinions, reasons, experiences).

**Strength:** you get deep, detailed answers and can ask follow-up questions ("Why do you say that?"). **Weakness:** it takes a lot of time, so you can only interview a few people.

#### 6.4.2 Surveys / Questionnaires

##### LEARN IT

A **survey** (or **questionnaire**) is useful when the **same set of questions** needs to be asked to **many people**.

- It may be on **paper** or created with **digital tools** such as **Google Forms** or **Microsoft Forms**.
- **Note:** online survey tools **automatically collect responses** and **display basic charts**, which makes data analysis much easier.

#### Types of survey and interview questions

| Question type                  | What it looks like                                                                       | Data it gives            |
|--------------------------------|------------------------------------------------------------------------------------------|--------------------------|
| <b>Closed: Yes/No</b>          | Do you study with your phone nearby? (Yes / No)                                          | Countable (quantitative) |
| <b>Closed: Multiple choice</b> | Which activity do you use your phone for most? (Lectures / Notes / Social media / Games) | Categories you can count |

|                       |                                                                         |                                            |
|-----------------------|-------------------------------------------------------------------------|--------------------------------------------|
| <b>Number / scale</b> | How many hours do you use your phone daily? ____                        | Quantitative                               |
| <b>Rating scale</b>   | "My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree | Quantitative (you can average the ratings) |
| <b>Open-ended</b>     | How does your phone help you in studies?                                | Qualitative (free text)                    |

## Good questions vs bad questions

A good question is **clear, short, unbiased** (it does not push the person towards one answer) and asks about **one thing at a time**.

| Problem                                               | Bad question                                          | Better question                                                                                                 |
|-------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Leading / biased</b> – suggests the "right" answer | Don't you agree that phones waste students' time?     | How do phones affect your study time? (Helps a lot / Helps a little / No effect / Harms a little / Harms a lot) |
| <b>Double-barrelled</b> – two questions in one        | Do you use your phone for lectures and games?         | Split into two: "Do you use your phone for lectures?" and "Do you use your phone for games?"                    |
| <b>Vague</b> – unclear words                          | Do you use your phone a lot?                          | How many hours do you use your phone on a normal school day?                                                    |
| <b>Overlapping options</b>                            | 1–3 hrs / 3–5 hrs / 5+ hrs (where does exactly 3 go?) | Less than 3 hrs / 3 to less than 5 hrs / 5 hrs or more                                                          |
| <b>Sensitive / too personal</b>                       | What is your father's monthly income?                 | Leave it out unless it is truly needed for the study.                                                           |

### WORKED EXAMPLE

**Building a short Google Form, step by step.** Research question: "How do Class XI students travel to college?"

**Step 1:** Open Google Forms → Blank form → Title: "Travel to College Survey". Add a description: "This survey is anonymous and for a class project. Do you agree to take part? (Yes/No)" – this is your **consent** question.

**Step 2:** Q1 (multiple choice): "How do you usually travel to college?" Walk / Bus / Rickshaw / Motorbike / Car / Other.

**Step 3:** Q2 (number): "How many minutes does your journey take?"

**Step 4:** Q3 (open-ended): "What is the biggest problem you face while travelling?"

**Step 5:** Click *Send* and share the link in the class group. Responses appear in the *Responses* tab with automatic pie/bar charts, and can be opened in Google Sheets.

**Result:** Q1 and Q2 give quantitative data; Q3 gives qualitative data.

### REMEMBER IT

**Interview = In-depth with few. Survey = Same questions to many.** Think of a doctor talking to one patient (interview) vs a census form sent to every house (survey).

### COMMON MISTAKE

**Mistake:** Writing only open-ended questions in a survey for 100 people. You will get 100 paragraphs that are very hard to analyse. **Fix:** Use mostly closed or number questions in a large survey, and only one or two open-ended questions.

### اردو میں سمجھیں

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔ سروے میں ایک ہی سوالات بہت سے لوگوں سے پوچھے جاتے ہیں، کاغذ پر یا گوگل فارمز (Google Forms) جیسے ڈیجیٹل ٹولز کے ذریعے، جو جوابات خودکار طور پر جمع کر کے چارٹ بھی دکھاتے ہیں۔ اچھا سوال واضح، مختصر، غیر جانبدار اور ایک وقت میں ایک ہی بات کے بارے میں ہوتا ہے۔

### EXAM TIP

You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the problem (leading, double-barrelled, vague, overlapping) and give the corrected version.

### CHECK YOURSELF

1. Name two digital tools for creating surveys.
2. What is wrong with: "Do you enjoy maths and physics?"
3. Are interview answers usually qualitative or quantitative?

---

Answers: 1. Google Forms and Microsoft Forms. 2. It is double-barrelled (two subjects in one question); ask about each subject separately. 3. Usually qualitative.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** An interview is:

- a. A spoken Q&A, often fewer people, deeper
- b. A 500-person tick sheet always
- c. A NAND
- d. A pie

**MCQ 2.** A survey is:

- a. The same questions to many people
- b. Always one celebrity chat
- c. A sort
- d. A cable

**MCQ 3.** Closed questions are easier to:

- a. Count and chart
- b. Explore feelings
- c. Build CPUs
- d. Route IP

**MCQ 4.** Open questions are better for:

- a. Reasons and quotes
- b. Instant averages
- c. Primary keys
- d. Gray codes

**MCQ 5.** A leading question is:

- a. Biased
- b. Required
- c. A foreign key
- d. A sprint

**MCQ 6.** Sampling matters because:

- a. Only asking prefects is not the whole college
- b.  $n$  always equals 1
- c. Charts ignore  $n$
- d. Ethics forbid samples

## Extra short questions

1. Differentiate interview and survey.
2. Two question-writing rules.
3. When mix both?
4. What is a pilot?

## Extra long question

1. Write 4 survey questions (2 closed, 2 open) on screen time, plus 3 interview prompts. Mark any bias you avoided.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

**Short**

1. Interview: deep, slow, few. Survey: standardised, many, shallower.
2. One idea per item; avoid leading; simple Urdu/English; pilot test.
3. Survey for numbers, then interview to explain a surprising number.
4. Try the form on 5 classmates and fix confusing items.

**Long**

1. Closed: hours bands; devices. Open: why night use; effect on sleep. Interview: walk me through yesterday evening. No "don't you think phones are evil".

---

Next lecture: 6.4.3 – 6.4.5 Prototypes, Observation, and Simulations